

Geopolitical Risk and Cross-Asset Implied Volatility: Evidence from Equity, Oil, and Gold Options

Danial Sanahurskyj*

Indiana University - Kelley School of Business

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Abstract

This paper examines how geopolitical shocks reprice risk across equity, oil, and gold options markets through both aggregate implied volatility and full volatility surface distortion. Using end-of-day options data on equities, gold, and oil and analyzing systematically identified geopolitical events through geopolitical newspaper coverage frequency, I document statistically significant but structurally distinct responses across the three asset classes. Equity volatility is uniform and risk is not priced into any specific region of the surface. Oil exhibits the most consistent surface distortion, concentrated in the deep out-of-the-money wings. Gold shows the largest response and responds more strongly to anticipated threats than realized acts. All three assets deflate after a geopolitical event consistent with the popular buy-the-rumor, sell-the-news mechanism. The most novel finding is that gold's structurally call-skewed surface intensifies sharply on event days, revealing that investors express safe-haven demand through upside call buying rather than downside put protection. Finally, shock severity above the identification threshold adds little explanatory power, consistent with geopolitical fear operating as a binary regime switch rather than a continuous variable.

*Supervised by Dr. Russell Rhoads and Dr. Karim Farroukh. Special thanks to Professor Lee Smales from the University of Western Australia for his feedback; this paper extends his work and draws substantial inspiration from it. For further inquiry, email: danial.sanahurskyj23@gmail.com, GitHub link with data and code: <https://github.com/DanialSanahurskyj/Geopolitical-Risk-and-Cross-Asset-Implied-Volatility-Code-and-Data>

1. Introduction

Geopolitical shocks reprice risk, but they do not reprice it uniformly. The same shock that can drive volatility to gold markets can leave equity markets unaffected. Prior work measures realized volatility, which is a backward-looking measurement, while implied volatility is forward looking: it embeds the market's expectation of future volatility (Christensen and Prabhala, 1998) and represents the price of insurance against future moves, so its response to a geopolitical shock reveals how investors expect to hedge what they fear. This study seeks to gain a deeper understanding of how geopolitical events reshape implied volatility in various asset classes and how investors seek option contracts to hedge geopolitical risk.

Previous research shows that external information shocks drive volatility and lead to negative investor sentiment. Major economic announcements are responsible for most of the observed intraday patterns of volatility in the markets (Ederington and Lee, 1993), and elections cause significant increases in volatility during the week of an election (Białkowski et al., 2008). Geopolitical conflict in the Middle East significantly increases volatility in oil and financial markets due to uncertainty and supply risks (Smales, 2021). Additionally, negative media sentiment is associated with increased stock market volatility and pessimistic investor behavior (Tetlock, 2007).

Researchers also suggest that gold and precious metals can be a viable option to hedge against geopolitical shocks (Baur and Smales, 2020), and that oil may provide diversification benefits against geopolitical and economic policy shocks (Kamal et al., 2022). Despite this body of evidence, existing studies focus on volatility as a single measure or focus on one asset class. None compares the effect of geopolitical events across various asset classes through both aggregate implied volatility and volatility surface distortion. To the best of my knowledge, no prior study has examined how geopolitical shocks simultaneously distort the full implied volatility surface, across moneyness and tenor, across equity, energy, and gold options markets within a unified cross-asset event study framework. This study aims to address that gap.

This study uses end-of-day close data on options from OptionMetrics from three asset class exchange traded funds, SPY (equities), IAU (gold), and USO (oil), spanning from January 1, 2010, to December 31, 2019. I identify geopolitical events systematically using the daily geopolitical risk (GPR) index of Caldara and Iacoviello (2022). The GPR index belongs to a

class of news-based uncertainty measures pioneered by Baker, Bloom, and Davis (2016), whose Economic Policy Uncertainty index established newspaper coverage frequency as a proxy for uncertainty; the GPR index applies this approach specifically to wars, terrorism, and interstate tensions. Days exceeding the 95th percentile of the index are classified as geopolitical events, resulting in 76 events during the sample period.

I find that geopolitical shocks produce statistically significant and structurally distinct implied volatility responses across the measured asset classes. Gold exhibits the largest surface distortion across events regardless of shock type, consistent with gold functioning as a universal safe-haven absorber of geopolitical uncertainty. Oil displays the smallest aggregate response but the most consistent surface distortion, concentrated in the deep out-of-the-money wings; following the event, oil implied volatility deflates below its pre-event baseline and remains suppressed, consistent with supply uncertainty resolving upon realization. Equities exhibit a uniform change in the surface, increasing volatility throughout the whole surface. All three assets revert to their pre-shock mean within the post-event window, suggesting markets pre-price anticipated geopolitical risk before the shock materializes and unwinds premium upon realization. Across the three assets, the implied volatility surface increases significantly in response to geopolitical shocks. Gold options exhibit a statistically significant change in skew, while equities and oil remain insignificant, indicating that geopolitical fear elevates option pricing differently across the three assets rather than uniformly.

Smales (2021) established that geopolitical risks drive volatility in oil markets and equity markets by analyzing standard deviations in returns of West Texas Intermediate crude oil, and the S&P 500 composite index, and Baur and Smales (2020) established gold correlated with geopolitical risks and gold acts as a hedge. This paper builds on these findings, by measuring aggregate implied volatility and volatility surface distortion across the combination of assets, using options data instead of returns. Studying the assets together through options surfaces reveals that equity, energy, and gold options markets price geopolitical uncertainty through structurally distinct channels, each with a characteristic implied volatility signature that is predictable from the nature of the shock.

The remainder of this paper is structured as follows: Section 2 describes the data and methodology used, Section 3 presents empirical results, and Section 4 concludes.

2. Data and Methodology

Data Sources

I use both aggregate implied volatility and implied volatility surfaces, with data from OptionMetrics. OptionMetrics is a widely used institutional database that stores and cleans options data across multiple variables, dates and assets. The data I use is available through The Wharton Research Data Services (WRDS), which partners with services like OptionMetrics to provide data to academics and researchers. Across the three asset classes, there are 10 years' worth of data, each containing a roughly equal split of put and call observations spanning January 1, 2010, to December 31, 2019. The IAU file is larger due to containing data going back to 2009, though this additional history was not used in my analysis. This study uses both put and call options to capture the full implied volatility surface, including smile asymmetry and directional skew. After filtering observations with missing data points, the files contain approximately 930 thousand observations for SPY, 924 thousand for USO, and 886 thousand for IAU, with missing values reflecting limited liquidity in very short-dated contracts during the sample period.

Asset Selection

The three assets chosen for this study reflect distinct geopolitical risk transmission channels. SPY, the SPDR S&P 500 ETF Trust, is chosen for having the most liquid S&P 500 options market with over 99% of S&P 500 ETF options going through SPY¹, capturing how geopolitical shocks transmit into macro uncertainty, growth expectations, earnings, and other financial conditions. USO, the United States Oil Fund, tracks West Texas Intermediate crude oil futures and proxies for supply shock risk, the channel through which geopolitical disruptions to oil-producing regions, energy infrastructure, and supply chains are priced into the market. Oil price shocks are not homogeneous; Kilian (2009) distinguishes supply shocks from aggregate demand and precautionary demand shocks, and geopolitical events operate primarily through the supply and precautionary channels that option markets must price in advance. IAU, the iShares Gold Trust, tracks spot gold prices and proxies for safe-haven demand, the property of an asset

¹ Ireland, C. J., Kotler, B. E., & Avila, A. (2026, January 23). *SPY liquidity: Flexibility to navigate any market*. State Street Investment Management. Not to be confused with S&P 500 futures contracts, where SPY represents over 99% of ETF based options.

that is uncorrelated or negatively correlated with other assets specifically during periods of market stress (Baur and Lucey, 2010), which investors seek during periods of geopolitical uncertainty (Baur and Smales, 2020). While GLD is the largest gold ETF, IAU was available through OptionMetrics and tracks the same underlying spot gold price, making it functionally equivalent for this analysis². ETFs are chosen over futures contracts because standardized options on ETFs provide consistent surface data through OptionMetrics. Together these three assets allow for a simultaneous cross-asset comparison of geopolitical fear across equity, energy, and safe-haven channels. To the best of my knowledge, no prior study has examined geopolitical fear transmission across these three channels using both aggregate implied volatility and volatility surface data.

Volatility Surface Construction

The implied volatility surface is a three-dimensional representation of implied volatility across two coordinates, moneyness and time to expiration. Following OptionMetrics standardized surface methodology, moneyness is expressed through the Black-Scholes delta (Black and Scholes, 1973) rather than the raw strike price. Delta measures sensitivity of an option's price to a one-dollar move in the underlying asset and serves as a natural cross-asset moneyness coordinate, as it approximates the probability of an option expiring in the money regardless of the underlying asset's price. This property makes delta-standardized surfaces directly comparable across SPY, USO, and IAU without requiring adjustment for differences in underlying strike price levels, a key advantage over strike-standardized surfaces for cross-asset analysis.

The put surface spans 7 delta points from $\Delta = -90$ to $\Delta = -10$ and the call surface spans 7 corresponding delta points from $\Delta = +10$ to $\Delta = +90$, with 5 tenor points ranging from 10 to 182 days to expiration. Both put and call options are included in this analysis to capture the full smile structure. The consistency of including both sides is grounded in put-call parity, which establishes a mechanical no arbitrage relationship between put and call prices at the same strike and tenor (Stoll, 1969):

² State Street Global Advisors, "SPDR Gold Shares (GLD) Annual Report 2020"; BlackRock, "iShares Gold Trust (IAU) Financial Statements 2020." At the conclusion of December 31, 2020, GLD managed approximately \$70.4 billion in assets, compared to IAU's \$29.1 billion.

$$C - P = S - Ke^{(-rT)}$$

Where C is the call price, P is the put price, S is the spot price of the underlying asset, K is the strike price, r is the risk-free rate, and T is the time to expiration. This relationship ensures that put and call implied volatilities at equivalent delta points contain complementary rather than redundant information about market pricing.

The standardized surface points are constructed by OptionMetrics using interpolation between observed option prices, a standard procedure in the academic options literature (Aït-Sahalia and Lo, 1998). Missing values arising from thin liquidity in very short-dated contracts are excluded from the analysis.

A central measure derived from the full surface is smile asymmetry, consistent with the volatility smile literature (Rubinstein, 1994), defined as the difference between put-side skew and call-side skew at the 30-day tenor:

$$ASYM = SKEW (put) - SKEW (call)$$

Where $SKEW (put) = IV(\Delta = -25, \tau = 30) - IV(\Delta = -50, \tau = 30)$ and $SKEW (call) = IV(\Delta = +25, \tau = 30) - IV(\Delta = +50, \tau = 30)$. A positive asymmetry indicates a put dominated surface where downside protection is priced at a premium relative to upside exposure, the standard condition for equity options. A negative asymmetry indicates a call dominated surface where upside demand exceeds downside protection pricing. This measure is central to identifying how each asset class prices geopolitical fear directionally.

Event Identification

In identifying geopolitical risks and events, I adopt the geopolitical risk (GPR) index of Caldara and Iacoviello (2022) to systematically measure the severity and magnitude of geopolitical events. Following Smales (2021) the GPR index is adjusted for publication timing and weekends, as an event that happens towards the end of the trading day or work week might not fully be priced into markets, an event occurring in a non-trading day is captured through newspaper coverage on the following trading day, ensuring alignment with index readings and market pricing³. Events are identified by counting the number of articles related to geopolitical

³ Of the 75 useable events, 5 fall on days the market was closed (3 federal holidays and 2 weekend days).

risk from numerous leading national and international newspapers. Published daily data can be found on Caldara and Iacoviello's website.⁴ To account for clustering of data and prolonged events, events occurring within seven calendar days of each other are grouped into a single cluster and represented by the peak GPR rating for the event and date. Using the daily GPR database and identifying days that exceed the 95th percentile threshold of 165.6, this study was able to identify 76 geopolitical events from January 1, 2010, to December 31, 2019, but because one of these events falls on January 1, 2010, and no prior pre-event data exists, this analysis uses 75 events. Some of these events are more covered, such as Brexit, Aramco Attacks and the Trump Election of 2016, others are more localized, such as Syria strikes, Iran sanctions and India-Pakistan Kashmir confrontation.

Unlike Smales (2021), who models volatility spillovers using continuous GPR changes and control variables such as VIX and EPU, I isolate discrete geopolitical shocks through event identification, making the pre-event baseline the natural control.

<Insert Figure 1>

<Insert Table 1>

Aggregate Implied Volatility

Prior to constructing full surface measures, aggregate implied volatility is reported for each asset at a standardized reference point, defined as delta equals negative 25 and tenor equals 30 days for put options. The 30-day horizon is consistent with the volatility horizon targeted by the CBOE VIX (Whaley, 2000), and the standardized point allows direct comparison across the three assets.

Abnormal Implied Volatility (AIV)

Using a similar methodology to MacKinlay (1997), abnormal implied volatility (AIV) is computed for each of the 75 geopolitical events identified earlier. While MacKinlay (1997) applies this framework to equity returns, this study adapts it to implied volatility surfaces, to the best of my knowledge the first application of event study methodology to cross-asset options surface data. AIV is computed by establishing a baseline implied volatility over the 20 trading

⁴ <https://www.matteoiacoviello.com/gpr.htm>

days prior to the event and measuring the peak implied volatility in the 10-day post-event window, expressed as a percentage deviation from the pre-event baseline. Using a 20-day pre-event window also helps avoid contamination from the event clustering. The AIV is found for all geopolitical events and then averaged out to find the mean change before and after an event. The formula for abnormal implied volatility is:

$$AIV = ([IV(\text{post-peak}) - IV(\text{pre-mean})] / IV(\text{pre-mean})) \times 100$$

The null hypothesis tested is that geopolitical events produce no abnormal implied volatility response, that is, AIV equals zero. AIV is expressed as a percentage to indicate the magnitude of implied volatility change attributable to the geopolitical event, with positive values indicating elevation above baseline and negative values indicating suppression below it. Results are robust to alternative baseline windows of 10 and 30 trading days as reported in the robustness section.

Surface Distortion Index

The Surface Distortion Index (SDI) is an aggregate measure of implied volatility surface distortion following a geopolitical event, complementing the point-specific AIV measure by capturing the mean absolute change across the full surface simultaneously. Each surface spans a put side and a call side, with each side composed of seven delta points crossed with five tenor points, yielding a 35-point grid per side and a combined 70-point grid across the full surface. Delta ranges from ± 10 to ± 90 , with tenors of 10, 30, 60, 91, and 182 days. SDI is computed using the AIV values defined in the previous section, taking the absolute value of the percentage deviation at each of the 70 surface points and averaging across the full grid. Absolute value is used in order to capture distortion magnitude rather than direction, a surface point that rises and one that falls equally represent equal distortion from baseline, and signed averaging would allow these to cancel, producing a misleadingly low SDI. The formula for the Surface Distortion Index is:

$$SDI = (1/N) \times \sum |AIV(i)|$$

Where $N = 70$ surface points and i indexes each delta-tenor combination across both puts and calls. A high SDI indicates that the geopolitical event caused substantial implied volatility distortion across the surface, while a low SDI indicates minimal surface-level disruption. SDI

enables direct cross-asset comparison of total surface distortion magnitude, allowing identification of which asset class absorbs the greatest geopolitical fear regardless of where on the surface that distortion concentrates.

Statistical Tests

To test the significance of the abnormal implied volatility response, a one-sample t-test is employed for each asset separately, testing whether the mean AIV across the identified geopolitical events is statistically distinguishable from zero. The null hypothesis is that geopolitical events produce no abnormal implied volatility response, that is, mean AIV equals zero. Statistical significance is reported at the one percent, five percent, and ten percent levels, denoted by three stars, two stars, and one star respectively throughout. The same framework is applied to the Surface Distortion Index, where a one-sample t-test for each asset tests whether mean SDI is significantly greater than zero, the null hypothesis being that geopolitical events produce no surface-level distortion.

Smile asymmetry is tested using a one-sample t-test against a null hypothesis of zero, a symmetric surface.

Following the sub-index framework of Caldara and Iacoviello (2022) and consistent with the approach of Baur and Smales (2020), events are further classified into act-driven and threat-driven subgroups using the GPR Acts and GPR Threats components. A t-test compares mean AIV between the two groups to test whether the nature of the geopolitical shock, realized event versus anticipated threat, produces systematically different surface responses across the three asset classes.

Finally, a Pearson correlation test examines whether GPR intensity above the identification threshold predicts the magnitude of the surface response, with a t-test on whether the correlation coefficient r is significantly different from zero. This tests the binary fear hypothesis that crossing the geopolitical risk threshold matters more than the severity of the shock above it.

Event Classification

Since not all geopolitical events are the same, a terrorist attack is structurally different from a diplomatic standoff; the 75 identified events are classified by type using the GPR Acts and GPR Threats sub-indexes provided by Caldara and Iacoviello (2022). The GPR Acts sub-index captures days dominated by realized geopolitical events such as attacks, invasions, and conflicts, while the GPR Threats sub-index captures days dominated by anticipated risk such as diplomatic tensions, military buildups, and political standoffs. Events are classified as act-driven when GPR Acts exceed 60% of the total GPR reading on the peak day, and as threat-driven when GPR Threats exceed 60%. Days where neither component exceeds 60% are classified as mixed events, where no single type of geopolitical risk dominated the coverage. This procedure yields 50 act-driven events, 9 threat-driven events, and 16 mixed events. Mixed events are retained in the full sample analysis but excluded from the act versus threat subgroup comparison. This classification extends the act versus threat framework of Baur and Smales (2020), who found that gold returns respond to geopolitical threats but not realized acts.

3. Empirical Results

Descriptive Statistics

Baseline implied volatility statistics at the reference point of delta equals negative 25, tenor equals 30 days, put options, reveal structurally distinct risk profiles across the three asset classes. SPY exhibits the lowest baseline implied volatility with a mean of 17.63% and a median of 16.32%, consistent with the relatively contained uncertainty typically priced into equity options over this period. The distribution is moderately right skewed at 1.683, with a maximum of 48.39% reached during the most acute stress episodes of the decade. USO shows substantially higher baseline implied volatility with a mean of 33.36% and a median of 31.54%, reflecting the persistent supply uncertainty embedded in energy options. The USO distribution is more symmetric than SPY at a skewness of 0.939 and excess kurtosis of 1.294, suggesting that elevated energy volatility is a routine feature of the market rather than a product of isolated extreme episodes. IAU presents a mean of 21.51% and a median of 19.89%. The IAU distribution is the most right skewed of the three at 2.223, with excess kurtosis of 11.193,

confirming the presence of anomalous surface episodes in gold options that are addressed in the robustness section.⁵

Over the sample period from January 1, 2010, to December 31, 2019, the daily GPR index averaged 93.29 with a standard deviation of 39.13, reflecting moderate but persistent geopolitical risk throughout the decade. The 95th percentile threshold used to identify geopolitical events was 165.63, with 184 spike days collapsing to 75 distinct events following the seven-day clustering procedure, of which 75 are usable, the largest GPR reading in the sample by a substantial margin. The second highest reading of 296.5 occurred on April 12, 2018, during the Syria Douma chemical attack and subsequent airstrike preparation period.

The pairwise correlations between daily implied volatility at delta equals negative 25 reveal that all three assets exhibit positive co-movement across the decade, though with meaningfully different magnitudes. SPY and USO exhibit the strongest co-movement at 0.471, reflecting shared sensitivity to broad macro stress. SPY and IAU co-move at 0.268, and USO and IAU at 0.227, both statistically significant at the one percent level. All three correlations are positive, indicating that geopolitical and macro stress tends to lift implied volatility across all three channels simultaneously, while the differentiation in magnitude establishes that each asset class nonetheless operates within a structurally distinct volatility regime.⁶

<Insert Figure 2>

<Insert Figure 3>

<Insert Table 2>

⁵ IAU's pre-event baseline IV is negatively correlated with its AIV response ($r=-0.31$, $p=0.008$): events with the lowest baseline IV produce a mean AIV of 29.05%, versus 16.97% for events with the highest baseline IV. This is the opposite of the leverage effect typical in equity options and is consistent with gold's safe-haven demand being acute when it has not been recently activated.

⁶ Daily IV levels are positively correlated across all three asset pairs (Figure 2), but this does not hold at the event level. IAU's AIV is uncorrelated with both SPY's ($r=-0.012$, $p=0.917$) and USO's ($r=0.031$, $p=0.795$), while SPY and USO remain correlated ($r=0.355$, $p<0.01$). IAU co-moves with the other assets under normal conditions but responds to geopolitical shocks through an independent channel.

Aggregate Implied Volatility Response and Window Response

Across all 75 identified geopolitical events, abnormal implied volatility responses are statistically significant for all three asset classes at the one percent level, confirming that geopolitical shocks produce measurable distortions in options pricing. However, the magnitude, direction, and persistence of these responses differ substantially across assets, consistent with the three-channel fear transmission framework proposed in this paper. To distinguish between the size of the response and the path through which volatility reprices, this section reports both peak abnormal implied volatility and the normalized event-window trajectory from 20 trading days before the event to 10 trading days after the event.

SPY exhibits a mean abnormal implied volatility response of 14.50% when measured at the peak post-event IV, with the response statistically significant at the one percent level ($t = 6.130$, $p < 0.001$). This confirms that equity options markets respond meaningfully to geopolitical shocks. However, the event-window path reveals a more counterintuitive dynamic. Before the event, SPY trades with a 20-day benchmark mean IV of 16.60% and a median of 16.30%. On the day before the event, SPY implied volatility is already 2.13% above its benchmark, consistent with anticipatory pricing in the days leading up to the GPR peak⁷. Immediately following the event, SPY implied volatility begins to deflate, falling below the pre-event baseline after the event and remaining suppressed through the tenth post-event day. This pattern is consistent with a buy-the-rumor, sell-the-news mechanism, whereby geopolitical uncertainty is priced into equity options during the buildup period and unwound once the event is realized, consistent with the established literature (Kelly, Pástor, and Veronesi, 2016).

IAU exhibits the largest abnormal implied volatility response among the three assets, with a mean AIV of 23.53% ($t = 6.657$, $p < 0.001$)⁸. This response is more than one and a half

⁷ SPY's response is also predicted by the direction of pre-event IV movement: events preceded by a rising 20-day trend produce a mean AIV of 19.35%, versus 9.77% for a falling trend ($t=2.07$, $p=0.042$). This extends the anticipatory pricing finding above, less of the repricing has occurred before the shock when IV has not yet begun to rise.

⁸ Two events (December 21, 2018 and January 2, 2019) had pre-event IAU baselines inflated by a Fed-driven volatility collapse — IAU's 30-day IV spiked to 91.69% on December 6, 2018 before falling to ~13% by month-end, unrelated to either event. Excluding these two events raises IAU's mean AIV from 23.53% to 26.27%, strengthening rather than weakening the headline result.

times the magnitude of SPY and roughly two and a half times the magnitude of USO, consistent with gold functioning as the strongest safe-haven absorber of geopolitical uncertainty. The event-window path further supports this interpretation. IAU trades in a more elevated volatility environment before geopolitical events, with a 20-day benchmark mean IV of 21.50% and a median of 21.00%. On the day before the event, IAU implied volatility is already above its baseline, suggesting that gold options begin pricing escalating uncertainty before the peak GPR reading is reached. After the event, implied volatility remains elevated for a couple of days before unwinding back down to below its baseline mean. On the first post-event day, IAU implied volatility sits significantly above baseline and remains above until after a couple days pass.

USO exhibits the smallest aggregate response, with a mean AIV of 9.49% ($t = 5.787$, $p < 0.001$), but its event-window behavior differs sharply from SPY. USO trades with a 20-day benchmark mean IV of 35.17% and a median of 34.91%. Prior to the event, implied volatility remains relatively subdued, with only modest evidence of anticipatory pricing. Following the event, USO implied volatility spikes briefly, captured by the peak-based AIV measure, before falling below its pre-event baseline and holding consistently under it through the tenth post-event day. Like SPY, USO unwinds volatility premium upon event realization; unlike SPY, the pre-event anticipation is modest, so the post-event suppression reflects the resolution of supply uncertainty rather than the unwinding of a pre-built risk premium.

Together, these results show that geopolitical shocks do not produce a uniform implied volatility response across asset classes. SPY reflects anticipatory pricing and post-event volatility deflation, consistent with uncertainty being resolved after realization. USO reflects a brief, wing-concentrated repricing followed by persistent suppression below baseline, consistent with supply uncertainty resolving once the shock materializes. IAU reflects the strongest and most persistent safe-haven response, with gold options remaining elevated before and after the event. Thus, aggregate AIV confirms that geopolitical shocks significantly affect option pricing, while event-window dynamics reveal the distinct transmission channels through which equity, energy, and gold markets absorb geopolitical fear.

<Insert Figure 4>

<Insert Table 3>

Cross-Asset Surface Distortion

Using the SDI to measure the impact of implied volatility across the full surface, which differs from aggregate IV by measuring 70 points across both the put and call surface rather than a single ATM reference point, all three assets exhibit highly significant surface distortion in response to geopolitical events.

SPY exhibits a mean SDI of 17.49% ($t=11.74$, $p<0.001$), indicating that geopolitical shocks produce statistically significant full-surface distortion in equity options markets. The high t-statistics reflect consistency of this distortion across events; the surface reliably moves in response to geopolitical shocks even if the direction varies.

USO shows a mean SDI of 23.28% ($t=16.77$, $p<0.001$) strikingly similar in magnitude to SPY despite representing a structurally different risk channel. The USO t-statistic of 16.77 is the strongest in the study, indicating that energy surface distortion is the most consistent of the three assets across events. USO produces meaningfully larger aggregate surface distortions than SPY, reflecting the concentration of energy fear in the deep OTM wings.

IAU exhibits by far the largest mean SDI of 56.40% ($t=11.71$, $p<0.001$) more than double the distortion of USO and more than three times that of SPY and leads surface distortion in many identified events regardless of shock type. IAU also has the highest standard deviation of the three assets with 41.43%, reflecting meaningful event to event variation in gold surface responses.

Surface Decomposition by Region

To analyze where distortion concentrates across the full implied volatility surface, mean AIV is reported across seven delta points and five tenor points for both put and call options, producing a complete 70-point surface decomposition matrix for each asset. At the 10-day tenor, extreme delta points at negative 10 and negative 90 for puts and positive 10 and positive 90 for calls carry elevated null rates of 12.6%, 20.2%, and 64.5% for SPY, USO, and IAU respectively, reflecting limited liquidity in very short-dated deep OTM contracts, and should be interpreted with caution.

For SPY, distortion is almost entirely explained by tenors. The surface inflates uniformly across all delta points at each tenor level, the 10-day tenor spikes between +17.1% and +36.2% across put deltas from -10 to -75, and between +17.7% and +53.6% across call deltas from +10 to +75, while the 182-day tenor sits in a narrow band around +5% on both sides. Delta barely matters for SPY; the surface inflates symmetrically. The one exception is the extreme deep wings, put delta -90 at 10d reaches +38.5% and call delta +90 at 10d reaches +54.3%, suggesting equity markets price a slight upside tail more than a downside tail during geopolitical events. Fear is short-lived and symmetric, consistent with equity markets quickly resolving geopolitical uncertainty into a known outcome.

For USO, the surface is asymmetric and concentrates in the deep OTM put wing. The put delta -75 at the 10-day tenor shows +104.0% while put delta -90 reaches +62.4%, confirming that crash protection demand spikes violently during geopolitical events. The call side is more moderate at ATM, call delta +50 at 10d is only +13.1% compared to put delta -25 at +26.7% but the deep OTM call wing also elevates with call delta +90 at 10d reaching +87.6%. Energy geopolitical fear therefore prices both extreme downside crash scenarios and extreme upside price spike scenarios, though with directional lean toward the put side at ATM. Importantly USO distortion persists into the back end, put delta -90 at 182d still shows +19.8% and call delta +90 at 182d shows +25.6%, consistent with energy supply uncertainty having a long horizon.

For IAU, the call side of the surface dominates the put side across nearly every tenor and delta combination. At the 10-day tenor, call delta +50 reaches +240.3% compared to put delta -50 at +149.6% the call ATM is 1.6 times more distorted than the put ATM. This asymmetry persists across all tenors: at the 182-day tenor, call delta +65 shows +65.3% while put delta -65 shows +58.3%. Gold implied volatility distortion does not decay meaningfully toward the back end, the 182-day call ATM still shows +27.2% indicating that safe-haven demand remains elevated for months, unlike equities, following geopolitical shocks. The IAU put delta -35 at the 10-day tenor shows an extreme reading of +359.2% driven by the January 2019 term structure inversion episode identified in the descriptive analysis and should be excluded from structural interpretation.

Together the three decomposition matrices confirm structurally distinct surface responses across the three transmission channels. SPY distortion is tenor-driven, symmetric, and short-lived

equity fear resolves quickly and prices uncertainty equally in both directions. USO distortion concentrates in the deep OTM wings with a bearish lean, energy fear prices extreme tail scenarios on both sides with more emphasis on downside. IAU distortion is call-dominated and persistent across all tenors, gold safe-haven demand expresses itself through upside call buying rather than downside put protection, a finding only visible through full surface analysis.

<Insert Figure 5>

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Smile Asymmetry and Skew

To measure the shape of the implied volatility surface, smile asymmetry is computed as the difference between put skew and call skew at the 30-day tenor, following the skew equation defined in the volatility surface construction section. This allows us to examine how market participants price directional risk during geopolitical uncertainty, whether they favor downside protection through buying or upside exposure through call buying. A baseline asymmetry is established across all 2,516 trading days in the sample and compared against asymmetry on the 75 identified geopolitical event days.

For SPY, the full-period asymmetry benchmark is +5.34pp, confirming that equity options are persistently put-dominated, OTM puts are systematically more expensive than OTM calls, consistent with the post-1987 equity smirk documented in the options literature (Rubinstein, 1994). On geopolitical event days the asymmetry is +5.22pp, a change of only -0.12pp that is not statistically significant ($t=-0.489$, $p=0.625$). The shape of the SPY surface remains virtually unchanged during geopolitical events. The surface inflates uniformly across the entire grid without tilting.

USO exhibits a full-period asymmetry of +2.40pp, more moderate than SPY and reflecting the two-sided nature of energy price risk. On geopolitical event days asymmetry falls slightly to +1.73pp, a change of -0.67pp that is not statistically significant ($t=-1.004$, $p=0.315$). Notably USO asymmetry turned negative on 259 trading days, 10.3% of the decade, indicating that on those days investors were pricing upside energy risk more than downside, consistent with the supply squeeze episodes identified in the descriptive analysis.

IAU presents the most striking finding. The full-period asymmetry of -11.82pp confirms that gold options are structurally call-skewed, OTM calls are persistently more expensive than OTM puts, meaning investors systematically seek upside gold exposure rather than downside protection. On geopolitical event days this call skew intensifies dramatically, with asymmetry falling to -24.14pp, a change of -12.32pp that is highly statistically significant ($t=-3.640$, $p<0.001$). When geopolitical events occur, investors in gold options significantly increase their demand for upside call options, amplifying the already call-skewed structure. This interpretation is consistent with Bollen and Whaley (2004), who demonstrate that net buying pressure directly shapes the implied volatility function, with demand concentrated in particular option series steepening the corresponding region of the surface. This finding directly extends Baur and Smales (2020), who documented that gold returns respond positively to geopolitical threats but did not identify the options-level mechanism, the surface analysis reveals that mechanism is upside call buying, not downside put protection.

<Insert Figure 7>

Act-Driven vs. Threat-Driven Events

Following the event classification framework established in Section 2, this section examines whether act-driven and threat-driven geopolitical events produce systematically different implied volatility surface responses across the three asset classes.

During act-driven events SPY produces a mean AIV of 14.90% ($t=4.977$, $p<0.001$) across all 50 act events, while threat-driven events produce a mean AIV of 9.71% ($t=1.574$, $p=0.154$) across the 9 threat events. The act-driven response is highly significant while the threat-driven response does not reach conventional significance levels, consistent with realized geopolitical shocks creating more immediate and measurable macro uncertainty than anticipated diplomatic tensions. The smaller threat-driven sample of 9 events limits statistical power and should be interpreted with caution.

USO follows a similar pattern. Act-driven events produce a mean AIV of 11.42% ($t=5.282$, $p<0.001$) while threat-driven events produce 5.17% ($t=1.156$, $p=0.281$), with the threat response again falling short of significance due to sample size. Realized geopolitical disruptions

to energy supply produce a stronger reaction than the mean AIV of 9.49%, showing that events tied to action cause larger oil market disruptions.

IAU shows the opposite pattern, responding more strongly to threat events at 39.72% ($t=2.879$, $p<0.05$) than act events at 23.57% ($t=5.766$, $p<0.001$). This is consistent with Baur and Smales (2020), who established that gold prices respond to anticipated geopolitical risk more than realized events, an expression of the buy-the-rumor, sell-the-news mechanism at the surface level. Gold investors position geopolitical uncertainty before it materializes rather than reacting after the fact.

Together these results reveal that the nature of geopolitical shock routes differently across the three transmission channels. Realized acts create more immediate and consistent uncertainty in equity and energy markets, while anticipated threats produce stronger safe-haven demand in gold. Further analysis is required to produce accurate interpretations of data as threat events are small in sample size. This divergence reinforces the three-channel fear transmission framework and directly extends the findings of Baur and Smales (2020) from returns data to the implied volatility surface.

<Insert Table 4>

GPR Intensity and Response Magnitude

Using GPR as a measure of geopolitical event intensity, this section examines whether the magnitude of a geopolitical shock above the identification threshold predicts proportionally larger implied volatility surface responses. This is distinct from the earlier finding that threshold-crossing events produce significant responses, the question here is whether more severe events produce proportionally more severe responses within the event sample.

The 75 identified events are divided into two groups using a GPR reading of approximately 240 (97th percentile) as the dividing line, separating high-intensity events such as the Paris attacks, Syria Douma, and the Qatar blockade from moderate-intensity threshold-crossing events, yielding 11 high-intensity and 64 moderate-intensity events. With only 11 high-intensity events, the sample size isn't large enough for an appropriate comparison. High-intensity events produce mean AIV responses of 12.1% for SPY, 13.1% for USO, and 16.7% for IAU. Moderate-intensity events produce 14.9%, 8.9%, and 24.7%. Notably, SPY and IAU show higher

mean AIV in the moderate-intensity group than in the high-intensity group, USO displays directionally positive scaling, though the correlation does not reach significance.

Formal Pearson correlation tests confirm this pattern. For aggregate AIV, SPY exhibits $r=-0.075$, USO $r=0.138$, and IAU $r=-0.063$, none statistically significant. Neither aggregate implied volatility nor full surface distortion shows a statistically significant relationship with GPR intensity above the identification threshold across any of the three asset classes.

This finding is consistent with geopolitical fear operating as a binary regime switch rather than a continuous scalar variable. Once a geopolitical event crosses the 95th percentile threshold of widespread media coverage, options markets reprice for the existence of the shock. How far above that threshold the GPR reading sits adds relatively little additional information, the market has already switched into risk-pricing mode regardless of whether the GPR reading is 170 or 354.

<Insert Table 5>

Robustness

To test whether the choice of pre-event baseline window affects the headline AIV findings, the primary 20-day baseline is replaced with alternative windows of 10 and 30 trading days. For SPY, mean AIV across the three windows is 14.23%, 14.50%, and 14.27% across 10, 20, and 30-day windows, showing near-perfect stability across specifications. USO likewise shows virtually no sensitivity, with mean AIV of 9.49%, 9.49%, and 9.79%. IAU shows the most variation across windows, with mean AIV increasing modestly from 22.12% at the 10-day baseline to 23.53% at the 20-day baseline and 24.35% at the 30-day baseline. This finding is most likely due to markets repricing uncertainty that builds up before an event. All nine specifications remain statistically significant at the one percent level, confirming that the headline AIV findings are robust to the choice of pre-event baseline window.

To assess whether the use of peak post-event implied volatility introduces upward bias into the AIV measure, the primary specification is rerun replacing peak post-event IV with mean post-event IV across the same 10-day window, such that the formula becomes:

$$AIV = ([IV(\text{post-mean}) - IV(\text{pre-mean})] / IV(\text{pre-mean})) \times 100$$

Under this alternative specification, SPY AIV falls from 14.50% to -0.79% ($t=-0.498$, $p=0.620$), USO from 9.49% to -0.03% ($t=-0.023$, $p=0.982$), and IAU from 23.53% to 1.99% ($t=0.732$, $p=0.467$), with all three results losing statistical significance entirely; this result confirms that geopolitical shocks produce a sharp, concentrated implied volatility spike that mean-reverts within the 10-day post-event window. The mean post-event measure captures primarily the reversion rather than the shock itself, validating peak post-event IV as the appropriate measure for this study and confirming that the effect is acute rather than persistent.

To assess sensitivity to the GPR event identification threshold, the full analysis is rerun at the 90th percentile threshold of 143.43, yielding 105 identified events and 103 usable events, compared to 75 at the primary 95th percentile specification. SPY mean AIV increases modestly from 14.50% to 15.37%, and IAU from 23.53% to 24.27%, with both remaining highly significant at the one percent level. USO shows the largest increase, rising from 9.49% to 13.99%, consistent with the lower threshold capturing a broader set of moderate-intensity supply-risk episodes that are more directly relevant to energy markets than to equities or gold. The ranking order of gold being the highest volatility asset and oil the lowest is still preserved under the alternative threshold, and all three assets remain significant at $p<0.001$, confirming that the three-channel fear transmission findings are not an artifact of the 95th percentile threshold choice.

4. Concluding Remarks

This paper examines how geopolitical events distort investor sentiment by analyzing implied volatility through three distinct channels: equities proxied by SPY, oil proxied by USO, and gold proxied by IAU. By systematically identifying 75 geopolitical events using the Caldara and Iacoviello (2022) GPR index at the 95th percentile threshold, this study analyzes implied volatility through aggregate abnormal volatility, smile asymmetry, volatility surfaces, and their relationship to geopolitical shocks across the period January 1, 2010, to December 31, 2019.

Equity options markets exhibit a counterintuitive deflationary volatility path following geopolitical events. With a statistically significant mean AIV of 14.50%, implied volatility falls below the pre-event baseline immediately following the event and remains suppressed through the full post-event window, consistent with geopolitical uncertainty being pre-priced and unwound upon realization. This pattern is consistent with the buy-the-rumor, sell-the-news

mechanism previously identified in gold returns by Baur and Smales (2020). The volatility surface inflates uniformly across all delta and tenor points without concentrating in a specific region, showing that investors broadly buy options in the presence of geopolitical uncertainty rather than hedging directionally.

Oil options markets exhibit the smallest aggregate response, with a mean AIV of 9.49% driven by a short-lived spike, after which implied volatility deflates below the pre-event baseline and remains suppressed throughout the post-event window, consistent with supply uncertainty resolving upon event realization. The surface response at the peak concentrates in the deep OTM wings on both sides, pricing extreme downside crash scenarios and upside price spike scenarios, and this wing distortion is the most statistically consistent response of the three assets across events.

Gold options markets exhibit the largest and most persistent response with a mean AIV of 23.53%. Implied volatility spikes sharply above baseline in the immediate post-event period and remains slightly elevated for a couple of days during the post-event period, reflecting persistent safe-haven demand following geopolitical shocks. The most novel finding of this paper is that the gold implied volatility surface is structurally call-skewed, OTM calls are persistently more expensive than OTM puts, and this call skew intensifies dramatically on geopolitical event days, with smile asymmetry falling from -11.82pp to -24.14pp , a change that is highly statistically significant ($t=-3.640$, $p<0.001$). This finding reveals that investors seek upside call exposure rather than downside put protection when pricing geopolitical uncertainty in gold, a mechanism only visible through full surface analysis.

Across all three assets, this paper finds that the magnitude of a geopolitical shock above the 95th percentile threshold adds little predictive power over the implied volatility response. GPR intensity and AIV correlations are near zero and statistically insignificant for all three assets. This binary fear hypothesis, that threshold crossing matters more than severity above it, holds under an alternative GPR threshold at the 90th percentile, suggesting that options markets switch into risk-pricing mode upon event realization rather than scaling their response proportionally to shock severity.

This paper is subject to several limitations. The sample period ends in December 2019, excluding major subsequent events including the COVID-19 pandemic and the Russia-Ukraine

conflict. The GPR index is constructed from media coverage, which means the event sample captures media-defined geopolitical risk rather than market-defined risk, and some identified events may not map cleanly onto market-moving shocks, such as the Osama bin Laden death announcement. The use of ETF options as proxies introduces basis risk relative to direct futures contracts, and IAU implied volatility at the $\Delta = -50$ reference point is affected by OptionMetrics interpolation from 2016 onward, addressed through the substitution of $\Delta = -25$ as the aggregate reference point throughout.

Future research could extend this framework by applying the cross-asset surface methodology to post-2020 and pre-2010 data to test whether the three-channel transmission structure holds under structurally different volatility environments. Expanding the asset universe to include currencies, credit spreads, and other commodity classes would allow a more complete mapping of how geopolitical shocks route through financial markets. Intraday options data would allow measurement of how quickly each channel reprices in the hours immediately following a shock, complementing the end-of-day framework developed here.

Geopolitical shocks reprice risk, but not uniformly. This paper shows that the non-uniformity is not random; it is systematic and predictable from the nature of the transmission channel. Understanding these distinct signatures allows investors to better anticipate how geopolitical shocks will affect the market depending on which channel they activate.

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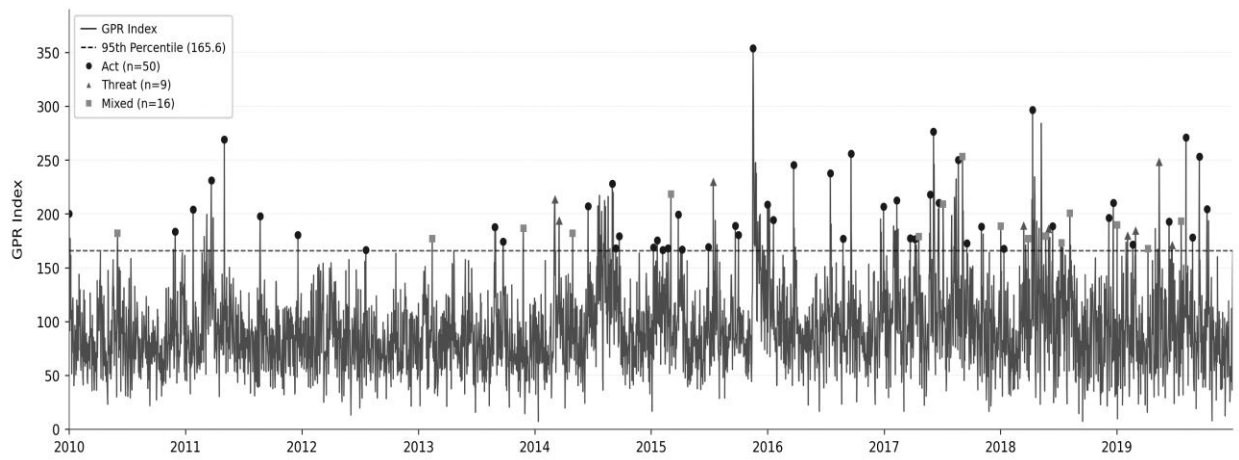
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Figure 1: GPR Index Timeline



Note: Event 1 was not used in this study to analyze volatility due to the lack of data prior to the event.

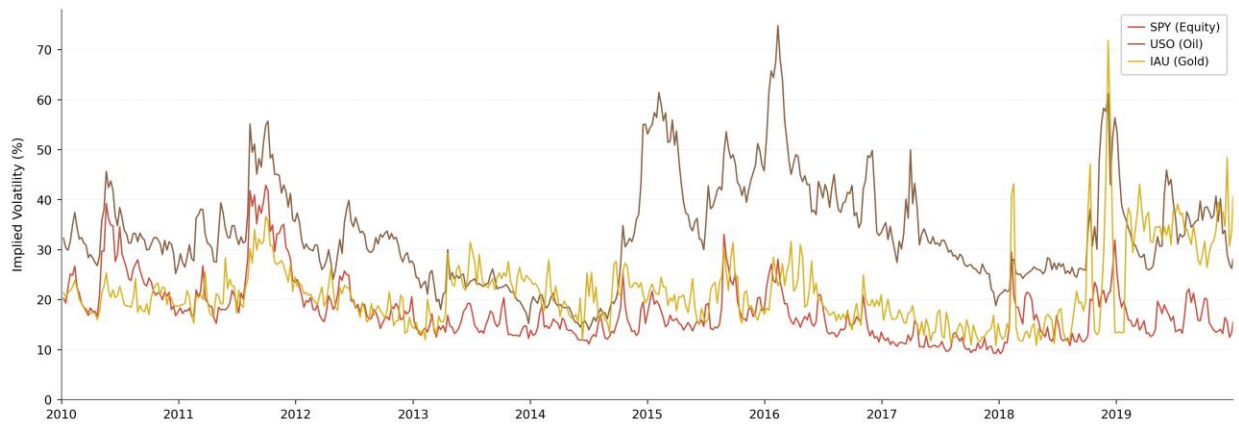
95th percentile: 165.6

Period: January 1, 2010 – December 31, 2019

Figure 2: Correlation of Implied Volatility in the Sample

	SPY IV	USO IV	IAU IV
SPY IV	1.0000	0.4713	0.2683
USO IV	0.4713	1.0000	0.2274
IAU IV	0.2683	0.2274	1.0000

Figure 3: Aggregate Implied Volatility for full Timeframe



Note: Chart uses delta of -25 and expiration of 30 days to measure implied volatility.

Figure 4: Abnormal Implied Volatility 20-days before an event and 10-days after

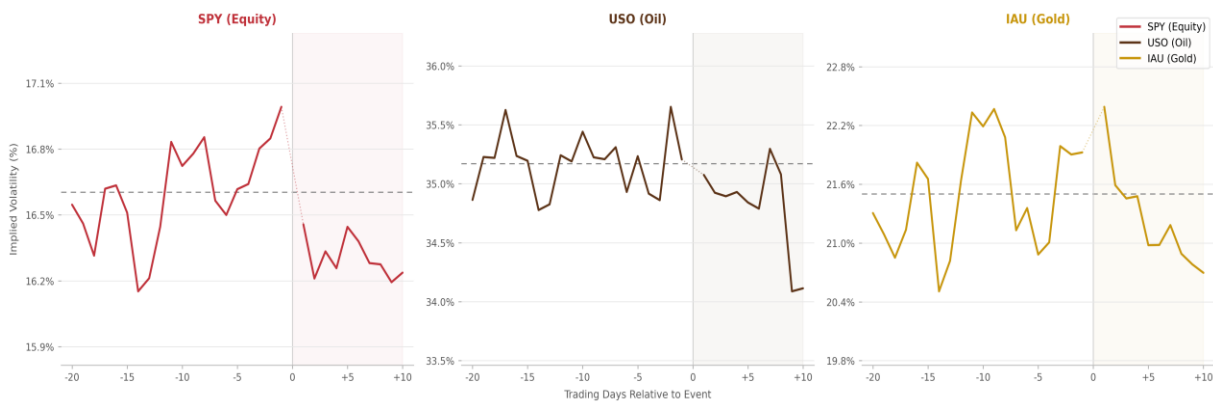
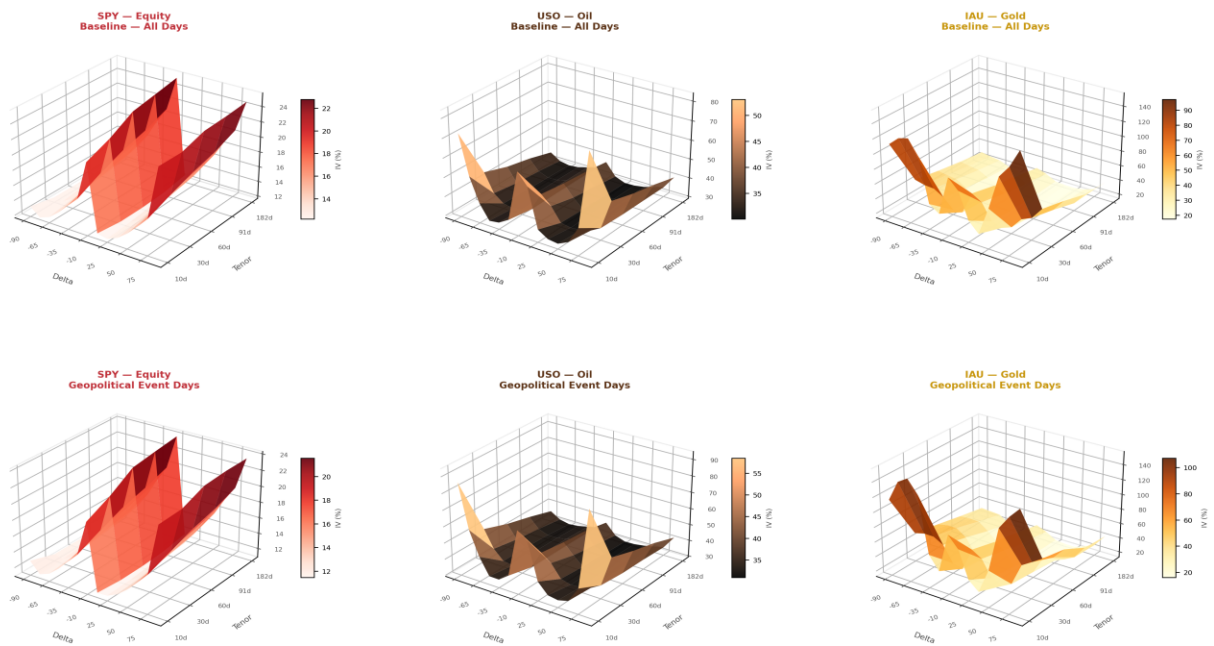


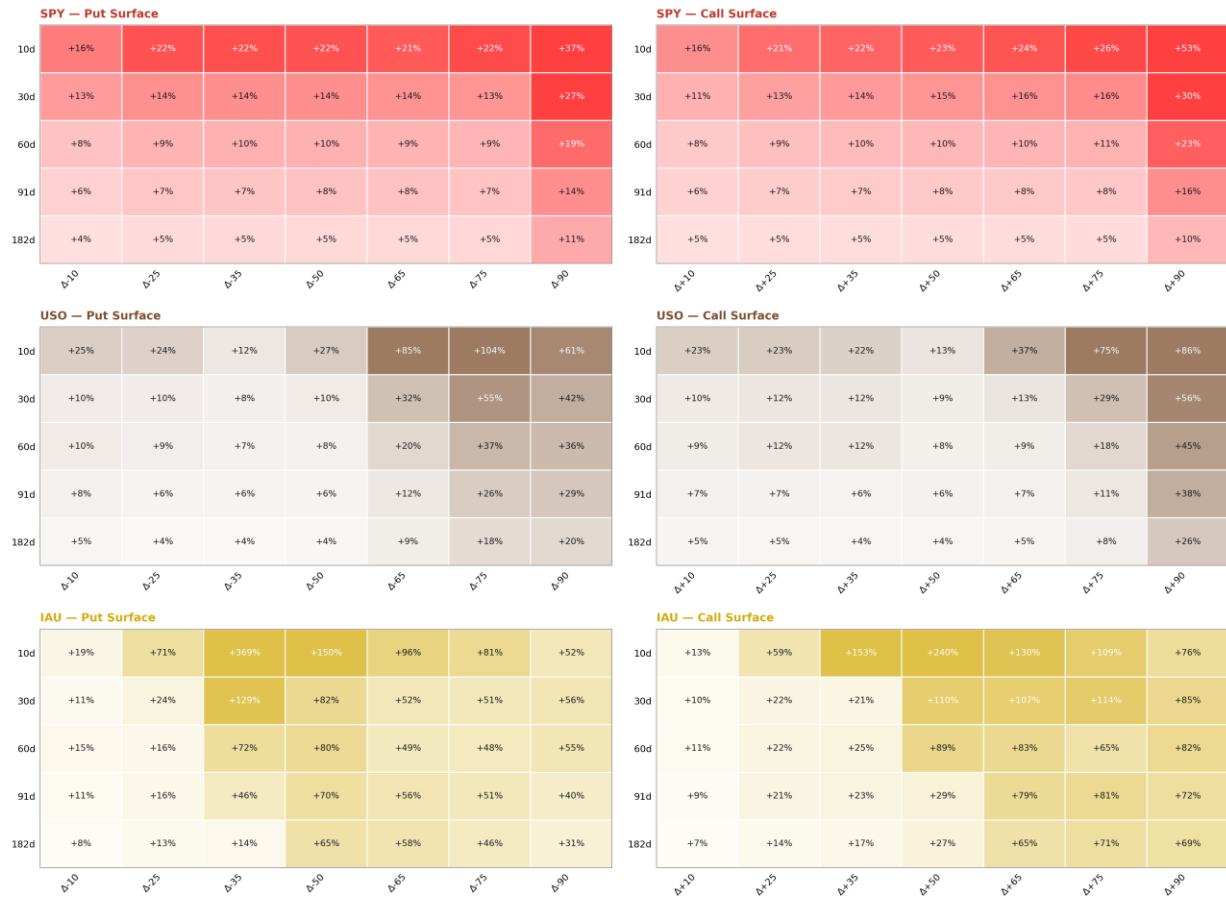
Figure 5: Volatility Surfaces (Puts and Calls)



Note: Implied volatility surface, delta standardized from $\Delta-90$ to $\Delta+90$, 5 tenors from 10 to 182 days

Top row: full-decade surface – Bottom row: mean surface on 75 geopolitical event days

Figure 6: Volatility Surface Decomposition



Note: Mean AIV (%) across 75 geopolitical events, delta-standardized surface, 20-day pre / 10-day post window.
Color intensity proportional to distortion magnitude, darker = larger distortion

Figure 7: Volatility Curve Smiles

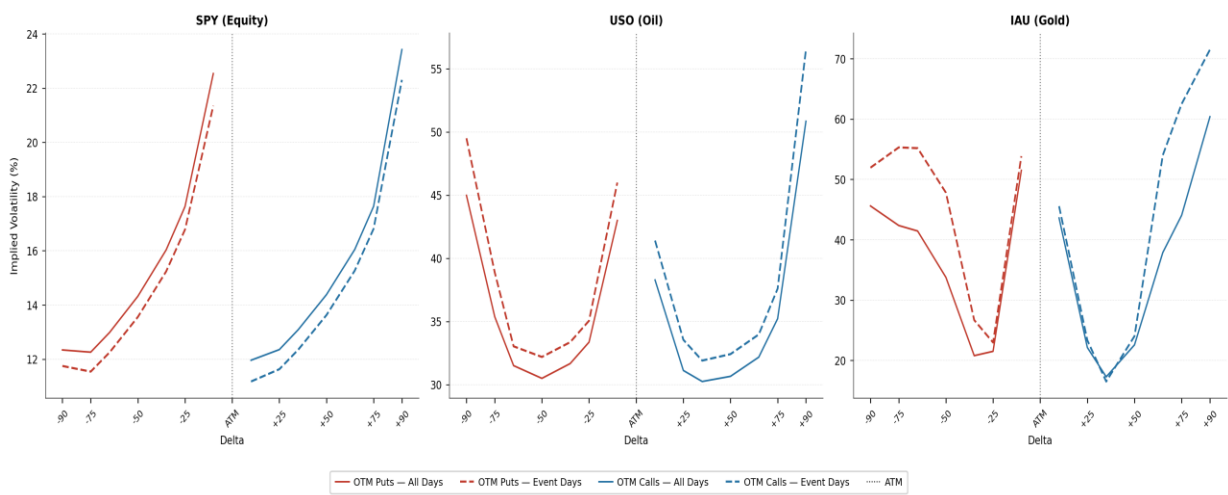


Table 1: Geopolitical Events by Year

Year	Events	ACT	THREAT	MIXED	Mean GPR	Max GPR	Min GPR
2010	3	2	0	1	188.5	200.2	181.9
2011	5	5	0	0	216.5	269.2	180.2
2012	1	1	0	0	166.4	166.4	166.4
2013	4	2	0	2	181.3	187.6	174.1
2014	7	4	2	1	196.0	227.8	168.2
2015	12	10	1	1	198.8	354.1	166.4
2016	7	7	0	0	217.9	255.7	177.0
2017	12	9	0	3	210.3	276.6	172.8
2018	12	5	2	5	196.1	296.5	167.5
2019	13	6	4	3	200.4	270.9	167.5
Total	76	51	9	16	201.4	354.1	166.4

Note: This table reports identified geopolitical events from January 1, 2010, to December 31, 2019, using daily GPR data from Caldara and Iacoviello's website. GPR data measures newspaper frequency coverage through specific geopolitical related words. Events are then classified by using the 95th percentile as a threshold. Events are subcategorized into acts versus threats by seeing which type of news drove 60% of the GPR, events with less than 60% are identified as mixed events. The event identified on January 1, 2010 (act-driven) is included in this table but excluded from all analysis due to the absence of pre-event data, so the analysis sample contains 75 events (50 act-driven). Full excel file of identified geopolitical events is on the GitHub page, with the link on the first page.

Table 2: Aggregate Implied Volatility for full Timeframe

	N	Mean	Median	Std. Dev.	Skewness	Kurtosis	Min	Max
<i>SPY</i>	2,516	17.63%	16.32%	6.07%	1.683	3.733	8.93%	48.39%
<i>USO</i>	2,516	33.36%	31.54%	10.72%	0.939	1.294	13.36%	96.78%
<i>IAU</i>	2,516	21.51%	19.89%	7.81%	2.223	11.193	9.85%	93.51%

Note: This table reports summary statistics for daily aggregate implied volatility for SPY (equities), USO (oil), and IAU (gold) over the sample period January 1, 2010, to December 31, 2019. Aggregate implied volatility is measured at the standardized surface point of delta = -25 and 30 days to expiration using put options, expressed in annualized percentage terms. N denotes the number of daily observations. Mean, Median, Std. Dev., Min, and Max are in percentage points; Skewness and Kurtosis are not percentage points. Implied volatility data are sourced from OptionMetrics via Wharton Research Data Services.

Table 3: Skew and Smile Data

	Put Skew	Call Skew	All Days Asymmetry	Event Days Asymmetry	Change (pp)	t-stat	p-value
<i>SPY</i>	+3.310pp	-2.027pp	+5.337pp	+5.221pp	-0.117pp	-0.489	0.6251
<i>USO</i>	+2.866pp	+0.465pp	+2.402pp	+1.731pp	-0.671pp	-1.004	0.3156
<i>IAU</i>	-12.263pp	-0.445pp	-11.818pp	-24.135pp	-12.316pp	-3.640	0.0003***

Note: This table reports volatility smile asymmetry for SPY (equities), USO (oil), and IAU (gold) over the sample period January 1, 2010 to December 31, 2019. All values are at the 30-day tenor and expressed in percentage points (pp). Put Skew is defined as $IV(\Delta = -25) - IV(\Delta = -50)$ and Call Skew as $IV(\Delta = +25) - IV(\Delta = +50)$. Asymmetry is the difference between put and call skew, with positive values indicating a put-dominated (downside-protective) surface and negative values a call-dominated (upside-demand) surface. All Days Asymmetry is the mean across all 2,516 trading days; Event Days Asymmetry is the mean across the 75 geopolitical event days. Change (pp) is the difference between event-day and full-period asymmetry, tested against a null of zero using a one-sample t-test. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

Table 4: Acts versus Threats Driven Geopolitical Events

	Act-Driven Events (n=50)				Threat-Driven Events (n=9)			
	N	Mean AIV	t-stat	p-value	N	Mean AIV	t-stat	p-value
<i>SPY</i>	50	14.90%***	4.977	<0.001	9	9.71%	1.574	0.154
<i>USO</i>	50	11.42%***	5.282	<0.001	9	5.17%	1.156	0.281
<i>IAU</i>	50	23.57%***	5.766	<0.001	9	39.72%**	2.879	0.021

Note: This table reports mean Abnormal Implied Volatility (AIV) for SPY (equities), USO (oil), and IAU (gold), separately for act-driven and threat-driven geopolitical events over the sample period January 1, 2010 to December 31, 2019. AIV is the percentage deviation of peak implied volatility in the 10-day post-event window from the mean implied volatility over the 20-day pre-event window, measured at $\Delta = -25$ and 30 days to expiration. Events are classified as act-driven when the GPR Acts sub-index exceeds 60% of the total GPR reading on the peak day, and as threat-driven when the GPR Threats sub-index exceeds 60%; mixed events, where neither component dominates, are retained in the full-sample analysis but excluded from this comparison. N denotes the number of events in each group. Mean AIV is expressed in percentage deviation and tested against a null of zero using a one-sample t-test. Given the small number of threat-driven events ($n = 9$), threat-group results should be interpreted with caution. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

Table 5: Response Magnitude to Geopolitical Events

	High-Intensity Events (n=11)			Moderate-Intensity Events (n=64)			GPR Correlation	
	Mean AIV	t-stat	p-value	Mean AIV	t-stat	p-value	r	p-value
<i>SPY</i>	12.14%	2.147	0.057	14.90%***	5.711	<0.001	-0.075	0.523
<i>USO</i>	13.07%**	3.328	0.008	8.88%***	4.931	<0.001	0.138	0.238
<i>IAU</i>	16.65%***	4.630	0.001	24.71%***	6.048	<0.001	-0.063	0.593

*Note: This table reports response magnitude to geopolitical events for SPY (equities), USO (oil), and IAU (gold) over the sample period January 1, 2010 to December 31, 2019. Events are split by intensity at a GPR reading of approximately 240, yielding 11 high-intensity and 64 moderate-intensity events. Mean AIV is the percentage deviation of peak implied volatility in the 10-day post-event window from the mean over the 20-day pre-event window, measured at $\Delta = -25$ and 30 days to expiration, expressed in percentage deviation and tested against a null of zero. The GPR Correlation columns report the Pearson correlation (r) between GPR intensity and Mean AIV across all 75 events, with the p-value testing whether r differs from zero. Given the small number of high-intensity events ($n = 11$), high-intensity results should be interpreted with caution. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels respectively.